

A Harmonic-Mixer-Based Fractional- N PLL Employing Voltage-Domain Feed-Forward Noise Cancellation

Haoming Zhang^{ID}, Graduate Student Member, IEEE, Masaru Osada^{ID}, Member, IEEE, Yuyang Zhu, Graduate Student Member, IEEE, and Tetsuya Iizuka^{ID}, Senior Member, IEEE

Abstract—A harmonic-mixer (HM)-based fractional- N phase-locked loop (PLL) employing voltage-domain feed-forward noise cancellation (FFNC) is presented in this article. By adding the output of the first-stage phase detector (PD) to that of the second-stage PD, the noise and power overhead from the first-stage voltage-controlled oscillator (VCO) can be suppressed, without relying on blocks such as high-speed delay lines that need to operate with a much higher bandwidth than the noise component that is being canceled. Architectural choices are made to mitigate the effect of process variation on the noise cancellation to avoid the need for calibration. A proof-of-concept prototype is implemented in 65-nm CMOS technology that achieves 106-fs rms jitter and -63 -dBc worst case fractional spur, and the phase noise (PN) improvement due to the proposed noise cancellation technique is demonstrated through measurement results.

Index Terms—Dual-feedback, feed-forward noise cancellation (FFNC), fractional- N phase-locked loop (PLL), harmonic-mixer (HM), PLL.

I. INTRODUCTION

THE suppression of quantization noise and spurs has been one of the main challenges in the development of fractional- N phase-locked loops (PLLs). Architectures based on quantization-error-cancellation (QEC), either in the time domain using the digital-to-time-converters (DTCs) or in the voltage domain using a digital-to-analog-converter (DAC), have been popular in recent years, with many of the state-of-the-art works in terms of jitter-power performance being based

Received 17 July 2024; revised 26 September 2024 and 10 November 2024; accepted 8 December 2024. Date of publication 17 December 2024; date of current version 25 July 2025. This article was approved by Associate Editor Jaehyoun Choi. This work was supported in part by the Institute for AI and Beyond, The University of Tokyo; in part by Japan Society for the Promotion of Science (JSPS) KAKENHI under Grant JP21H03406 and Grant JP21J21917; in part by Japan Science and Technology Agency (JST)-Mirai Program under Grant JPMJMI20A1; and in part by Toshiba Electronic Devices and Storage Corporation. (Haoming Zhang and Masaru Osada are co-first authors.) (Corresponding author: Tetsuya Iizuka.)

Haoming Zhang and Yuyang Zhu are with the Department of Electrical Engineering and Information Systems, The University of Tokyo, Tokyo 113-0032, Japan (e-mail: zhm@g.ecc.u-tokyo.ac.jp; zhuyuyang@silicon.u-tokyo.ac.jp).

Masaru Osada was with the Department of Electrical Engineering and Information Systems, The University of Tokyo, Tokyo 113-0032, Japan. He is now with Apple, Inc., Tokyo 106-6108, Japan (e-mail: osada@silicon.u-tokyo.ac.jp).

Tetsuya Iizuka is with the Systems Design Laboratory, School of Engineering, The University of Tokyo, Tokyo 113-0032, Japan (e-mail: iizuka@vdec.u-tokyo.ac.jp).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JSSC.2024.3516139>.

Digital Object Identifier 10.1109/JSSC.2024.3516139

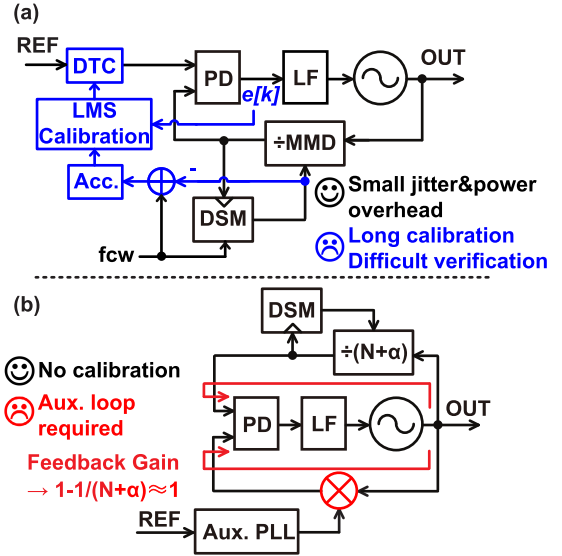


Fig. 1. Basic structure of (a) DTC-based PLL and (b) dual-feedback HM-based PLL.

on this architecture [1], [2], [3], [4]. However, the circuits used for the QEC are unavoidably affected by PVT-related gain errors and non-linearity, requiring intensive digital calibration to prevent severe performance degradation. For example, in a DTC-based PLL, as shown in Fig. 1(a), an imperfect QEC can cause residual time error at the PD input, resulting in large in-band fractional spurs at near-integer channels, which cannot be filtered by the PLL loop [3]. Therefore, the characteristics of the DTC must be corrected using gain and linearity calibration schemes such as the least-mean-square (LMS) algorithm so that the DTC delay can accurately match the time-domain Q-error. These background calibrations, which can take hundreds of microsecond [3] or tens of millisecond [4] to converge, have to be performed each time the PLL switches to another frequency. This results in a significantly longer settling time compared to the PLLs that suppress the quantization noise inherently with the filtering effect of the loop [5], [6], [7], [8], where the settling process consists of only the locking transient. The frequency agility of the PLL, however, is stringently required in many applications such as wireless transceivers, where the PLL is used as the signal generator and its slow settling behavior can have a direct negative impact on the overall system performance [9], [10], [11], leading

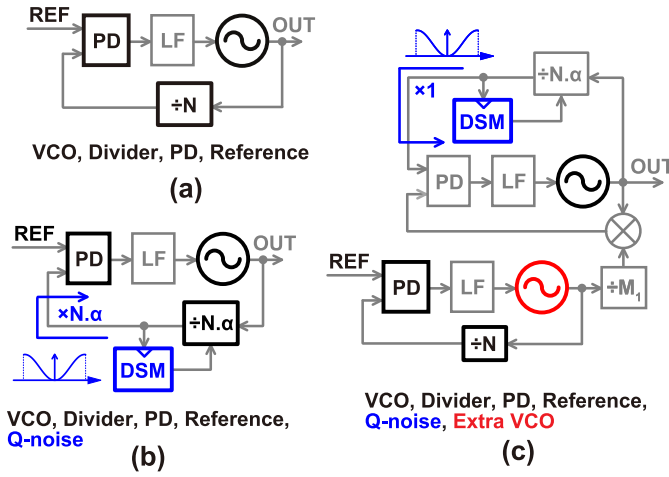


Fig. 2. Dominant sources of noise in (a) integer- N PLL, (b) fractional- N PLL, and (c) dual-feedback HM-based PLL.

to increased latency or higher power consumption. Another concern is that the residual quantization noise and fractional spurs resulting from QEC-based approaches are hard to model and predict because they result mainly from non-idealities in the circuit and second-order effects. In contrast, since the noise contribution of the quantization noise in a PLL without background calibration for the QEC is completely linear, it can be predicted based on an ideal linear model with general loop characteristics, and non-idealities in the circuit and second-order effects cause minor deviations from this prediction.

One of the effective methods to suppress the quantization noise and spurs without using the QEC is to employ a harmonic-mixer (HM)-based feedback [12], [13], [14], [15], [16]. As shown in Fig. 1(b), by employing a feedback path based on an HM in addition to the one based on a frequency divider seen in conventional PLLs, the total feedback gain is increased to almost unity, and the quantization noise and spurs avoid amplification in the loop [13]. As a result, the quantization noise and spurs are suppressed without relying on calibration-intensive techniques such as DTCs. The downside of this architecture is that at least one auxiliary PLL is required, potentially resulting in noise and power overhead that leads to a degradation in terms of jitter-power performance. Thus, the concept of noise cancellation, which is widely applied in low-noise amplifiers (LNAs) [17] and wireless transceivers [18], is introduced in our architecture, performing between two PLLs to mitigate the noise and power overhead from the auxiliary loop and overcome the overall performance limitation. It is worth noting that the proposed technique is not limited to HM-based PLLs but can also be applied to other cascaded PLL architectures to reduce the overhead from having multiple PLLs.

The rest of this article is organized as follows. In Section II, we discuss the noise and power overhead issues in the HM-based PLL and some ideas from prior arts that can be used to mitigate it. In Section III, we present the proposed architecture and explain its benefits and characteristics. We also analyze the noise transfer characteristics in the proposed architecture and the effects of gain error due to

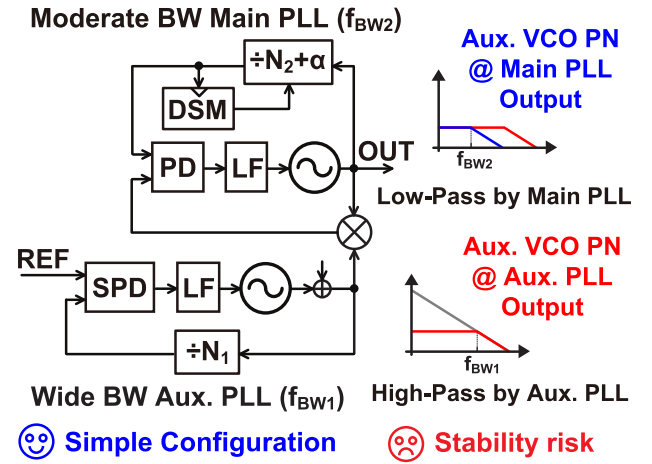


Fig. 3. Use of a wide bandwidth auxiliary PLL to suppress the auxiliary VCO PN.

process variation. Then, in Section IV, we explain the details of the circuit implementation and show simulation results to demonstrate the robustness of the proposed circuit over PVT variations. Section V summarizes the measurement results. Finally, we conclude this article in Section VI.

II. NOISE AND POWER OVERHEAD IN HM-BASED PLLS

In this section, we first explain the issue for the dual-feedback HM-based PLL [13] of having the noise and power overhead from an extra voltage-controlled oscillator (VCO). Then, we look at some potential methods to mitigate the overhead, along with the issues associated with them.

Fig. 2 summarizes the noise overhead in the conventional fractional- N and HM-based fractional- N PLLs compared to an integer- N PLL. In a conventional integer- N PLL shown by Fig. 2(a), the total noise is dominated by the VCO noise, the PD, the divider, and reference noise. The noise from the loop filter (LF) is usually at a negligible level and is often ignored in the noise analysis. As for a fractional- N PLL in Fig. 2(b), the fractional operation brings about an additional quantization noise. In the HM-based PLL in Fig. 2(c), to suppress the quantization noise, the number of PDs, dividers, and VCOs are all doubled due to an extra PLL. However, the PD and divider noise of the main PLL are negligible compared to that of the auxiliary PLL because they are not amplified due to the unity dc gain of the main loop. The HM in the main loop can also be designed to have a negligible phase noise (PN) contribution with relatively low power consumption based on prior works [15]. As a result, with a greatly suppressed quantization noise, the noise and power overhead of the HM-based PLL compared to an integer- N PLL mainly comes from the extra VCO. Thus, this work aims to realize an HM-based PLL with reduced noise and power contribution from the extra VCO while retaining the benefits of quantization noise and fractional spur suppression.

A key point in suppressing the overhead from the extra VCO is that the auxiliary PLL PN experiences low-pass filtering from the main PLL. As shown in Fig. 3, the auxiliary VCO PN first gets high-pass filtered by the auxiliary PLL and then low-pass filtered by the main PLL. As such, the auxiliary VCO PN can be suppressed effectively as long as the auxiliary PLL

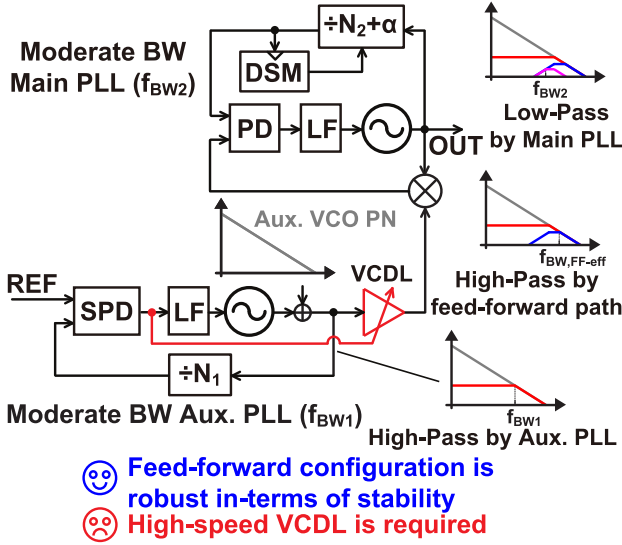


Fig. 4. Use of FFNC based on a VCDL to suppress the auxiliary VCO PN.

has a much wider bandwidth than the main PLL. However, the PLL bandwidth affects the loop stability [19] and is typically restricted to well below half of the reference frequency to ensure a sufficiently large phase margin, limiting the VCO PN suppression. Besides, the bandwidth limitation becomes even more severe when lower reference frequencies are used, which impose stricter limitations on the maximum achievable bandwidth, as a high-frequency reference can be impractical in terms of the system cost when generated by the crystal (XTAL) oscillator widely used in practice [15].

Another way to suppress the VCO PN without degrading the phase margin is to use feed-forward noise cancellation (FFNC) with a voltage-controlled-delay-line (VCDL) proposed in [20]. Fig. 4 shows the implementation of a moderate-bandwidth PLL with VCDL-based FFNC being used as the auxiliary PLL in an HM-based PLL. In this architecture, an auxiliary SPLL with a moderate or narrow bandwidth first generates the desired frequency signal. Then, a VCDL is placed at the SPLL output, and the output of the sampling phase detector (SPD) is used to control the VCDL. Since the SPD output contains the PN of the auxiliary PLL output converted to a voltage, the VCDL cancels the PN and generates a clean signal. Since the feed-forward path is independent of the PLL closed-loop system, the FFNC will not introduce any stability issues, while it can provide effective high-pass filtering for the VCO PN with a bandwidth up to half of the reference frequency [20]. However, the main downside of this method is the need for a high-speed VCDL. Since the VCDL is connected to the output of the PLL, it must operate at multiple GHz while also covering a sufficiently wide delay range to account for the skew and jitter, resulting in large noise and power overhead. Furthermore, the gain of the VCDL can be difficult to control over process variations, requiring complex calibration to ensure that the feed-forward path maintains the correct gain for effective cancellation.

III. PROPOSED ARCHITECTURE

Without the need for a VCDL, we proposed a voltage-domain FFNC to suppress the VCO PN, which will

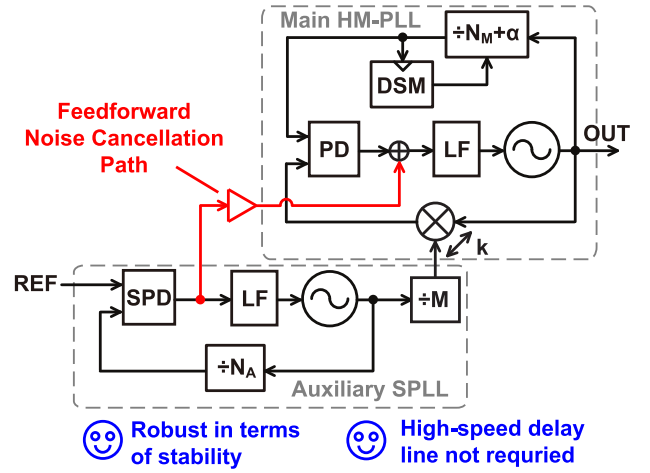


Fig. 5. Proposed FFNC concept.

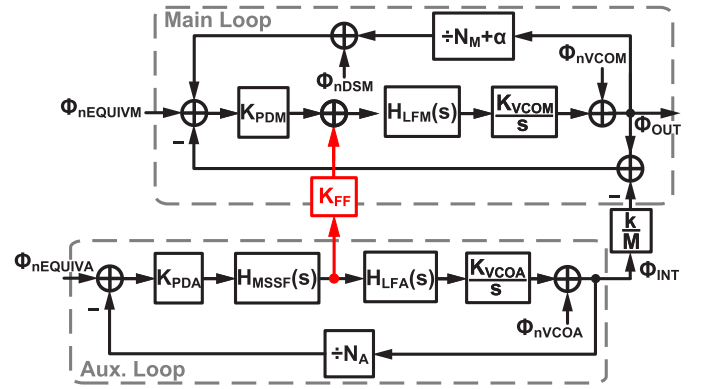


Fig. 6. Linear model of the proposed PLL with noise sources.

be detailed in this section. A conceptual block diagram of the proposed architecture is shown in Fig. 5, in which an SPLL is used as the auxiliary loop and the main loop uses an HM-based dual-feedback PLL. The output of the auxiliary SPD is directly added to the output of the main PD, providing the auxiliary VCO PN with wide bandwidth high-pass filtering as will be detailed in the following. Similar to the VCDL-based FFNC, the proposed FFNC has no impact on the loop characteristic nor stability of the auxiliary PLL, as the FFNC path is outside the auxiliary loop. It also does not affect the stability of the main PLL since it introduces no extra poles or zeros to the main PLL's closed-loop system.

A. Main PLL Noise Sources

Now, we take a more detailed look at the proposed concept by analyzing the linear model shown in Fig. 6. We first delve into the noise transfer function (NTF) from the noise sources in the main PLL to ϕ_{OUT} , which is the PN at the main PLL output. Initially, the open-loop gain of the main PLL is given by

$$H_{OLM}(s) = \frac{K_{PDM} K_{VCOM} H_{LFM}(s)}{s} \left(1 - \frac{1}{N_M + \alpha} \right) \quad (1)$$

where K_{PDM} and K_{VCOM} are the gains of the main PD and VCO, respectively. $H_{LFM}(s)$ is the transfer function of the

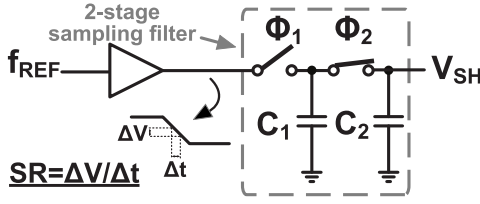


Fig. 7. Sampling PD in the auxiliary loop.

main LF, and $N_M + \alpha$ is the main PLL division ratio. The noise sources in the main PLL include the main VCO noise ϕ_{nVCO} , the quantization noise ϕ_{nDSM} , and $\phi_{nEQUIVA}$ as the total equivalent noise from the main divider, main PD, and HM referred to the main PLL input. The NTF of these noise sources to the main PLL output can then be calculated as follows:

$$H_{VCOM,OUT}(s) \equiv \frac{\phi_{OUT}}{\phi_{nVCOM}} = \frac{1}{1 + H_{OLM}(s)} \quad (2)$$

$$H_{DSM,OUT}(s) \equiv \frac{\phi_{OUT}}{\phi_{nDSM}} \approx \frac{H_{OLM}(s)}{1 + H_{OLM}(s)} \quad (3)$$

$$H_{EQUIVM,OUT}(s) \equiv \frac{\phi_{OUT}}{\phi_{nEQUIVM}} \approx \frac{H_{OLM}(s)}{1 + H_{OLM}(s)}. \quad (4)$$

We can see that $H_{VCOM,OUT}(s)$ is a high-pass characteristic similar to that in a conventional PLL. While $H_{DSM,OUT}(s)$ and $H_{EQUIVM,OUT}(s)$ are both low-pass characteristics with a dc gain of approximately 1, indicating that they are not amplified. The above analysis of the main loop is also consistent with our prior work, and further details can be referred to [13].

B. Auxiliary PLL Noise Sources

We then discuss the NTF of the auxiliary PLL noise sources, including ϕ_{nVCOA} as the noise of the auxiliary VCO and $\phi_{nEQUIVA}$ as the total equivalent noise from the reference, auxiliary divider, and auxiliary SPD referred to the reference input. The open-loop transfer function of the auxiliary PLL $H_{OLA}(s)$ is given by

$$H_{OLA}(s) = \frac{K_{PDA} K_{VCOA} H_{LFA}(s)}{N_A s} H_{SF}(s) \quad (5)$$

where K_{VCOA} is the gain of the auxiliary VCO and K_{PDA} is the gain of the SPD, which is given by [21]

$$K_{PDA}(s) = \frac{SR}{2\pi f_{REF}} \quad (6)$$

where SR is the slew rate of the sampling edge in the SPD, as shown in Fig. 7. $H_{LFA}(s)$ is the transfer function of the auxiliary LF, whose characteristic and definition will be discussed in detail in Section IV. N_A is the auxiliary PLL division ratio. In addition, $H_{SF}(s)$ is the transfer function of a two-stage sampling filter (SF) [22] used in the SPD in the auxiliary loop, which is given by

$$H_{SF}(s) = \frac{\sin(\pi f/f_{REF})}{\pi f/f_{REF}} e^{-s/2f_{REF}} \frac{1}{1 + \frac{C_2}{C_1 f_{REF}} s} \quad (7)$$

where C_1 and C_2 are the capacitors in the two-stage sampler of the SPD, as shown in Fig. 7, and f_{REF} is the reference frequency. The first term reflects the sampling operation, while the second introduces a delay or phase shift. The third term accounts for the charge-sharing effect between C_1 and C_2 . In addition, we define M as the division ratio of the intermediate divider placed between the auxiliary PLL and main PLL to avoid coupling and k as the harmonic index of the HM.

Based on the above derivation, as one of the key points of the proposed technique, the NTF from the auxiliary VCO noise to the main PLL output $H_{VCOA,OUT}(s)$ is given by

$$H_{VCOA,OUT}(s) \equiv \frac{\phi_{OUT}}{\phi_{nVCOA}} = \frac{1}{1 + H_{OLA}(s)} \frac{H_{OLM}(s)}{1 + H_{OLM}(s)} H_{FFNC}(s) \quad (8)$$

where

$$H_{FFNC}(s) = \left(\frac{k}{M} - \frac{K_{FF} K_{PDA}}{N_A K_{PDM}} H_{SF}(s) \right). \quad (9)$$

K_{FF} is the voltage gain of the feed-forward path, which will be detailed in Section IV. K_{PDM} equals $V_{DD}/2\pi$ as an XOR PD will be used in our case. It is not V_{DD}/π as in conventional cases because two divide-by-2 will be added before the XOR PD inputs to prevent spectrum folding of the DSM quantization noise [15]. We can see from (8) that $H_{VCOA,OUT}(s)$ consists of the high-pass filtering from the auxiliary PLL, the low-pass filtering from the main PLL, and a term $H_{FFNC}(s)$ provided by the proposed technique, corresponding to the difference between the gain of the main signal path and that of the noise cancellation path. If $H_{FFNC}(s)$ equals zero, the auxiliary VCO noise ϕ_{nVCOA} is completely canceled at the main PLL output. However, this is not achievable since $H_{FFNC}(s)$ is not a constant but contains a $H_{SF}(s)$, which reflects the sampling process of the SPD, preventing the detection of PN components whose frequencies are higher than $f_{REF}/2$. That being said, if the gains of the two paths match perfectly where

$$\frac{k}{M} = \frac{K_{FF} K_{PDA}}{N_A K_{PDM}} \quad (10)$$

and the filtering characteristic of the FFNC is given by

$$H_{FFNC}(s) = \frac{k}{M} (1 - H_{SF}(s)). \quad (11)$$

The characteristic of this $H_{FFNC}(s)$ is depicted by the black line in Fig. 8. Notably, it exhibits a high-pass characteristic with a cutoff frequency of close to $f_{REF}/2$. Therefore, even though perfect cancellation remains impossible, the proposed FFNC technique provides additional high-pass filtering for the auxiliary VCO noise, with a bandwidth close to half the reference frequency, which is much wider than conventional architectures. In Fig. 8, we have also shown the characteristics with different levels of the gain error between the main signal path and the noise cancellation path, where (10) is not fully satisfied. The results show that the gain error can lead to a degradation of the filtering effect at low frequencies. Fig. 9(a) shows the calculated PN spectrum of the overall output with

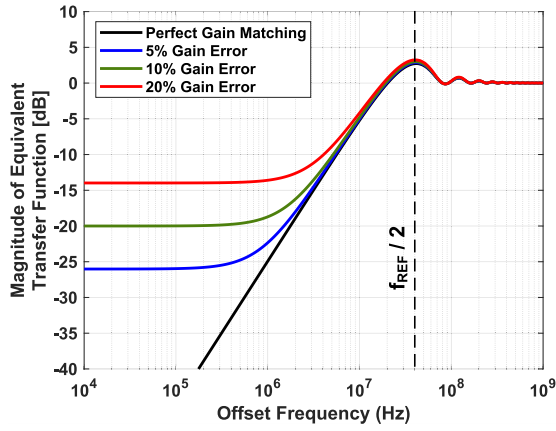


Fig. 8. Effective filtering characteristic $H_{FFNC}(s)$ on the auxiliary VCO PN for different values of the gain error.

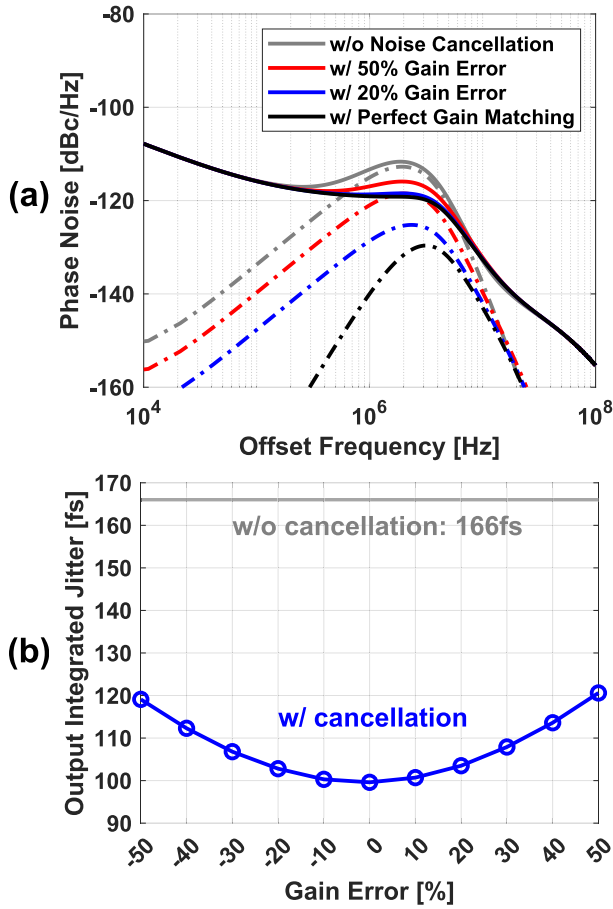


Fig. 9. (a) Calculated PN spectrum of the overall output and auxiliary VCO contribution and (b) integrated jitter for different gain error values.

auxiliary VCO contribution, and Fig. 9(b) shows the integrated jitter of the overall PLL for different values of the gain error. We can see from both figures that noise cancellation provides a significant improvement in noise compared to the case without noise cancellation. In addition, the effect of cancellation is almost the same regarding the sign of the gain error. At the same time, the performance degradation is negligible as long as the gain error can be maintained within $\pm 20\%$. This result shows that the proposed technique is robust to gain error. That

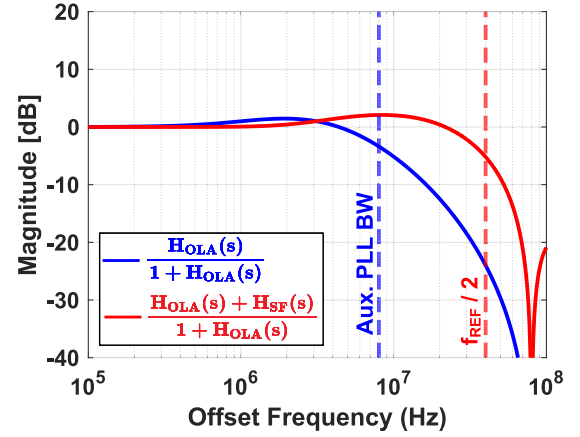


Fig. 10. Filtering effect of the auxiliary PLL with FFNC on the reference-referred noise.

being said, the gain of blocks such as the SPD can have a variation that is significantly larger than $\pm 20\%$ if we employ conventional techniques, requiring background calibration and reducing the benefit of the HM-based architecture. Therefore, we have employed some circuit techniques to suppress the gain error over process variation, as will be explained in Section IV.

Finally, we look at the NTF from the equivalent reference-referred noise $\phi_{nEQUIVA}$ to the overall output, denoted as $H_{EQUIVA,OUT}(s)$. For simplicity, supposing that (10) is satisfied, $H_{EQUIVA,OUT}(s)$ is given by

$$H_{EQUIVA,OUT}(s) \equiv \frac{\phi_{OUT}}{\phi_{nEQUIVA}} = \left(N_A \frac{H_{OLA}(s) + H_{SF}(s)}{1 + H_{OLA}(s)} \right) \frac{k}{M} \frac{H_{OLM}(s)}{1 + H_{OLM}(s)}. \quad (12)$$

We can see that it contains three terms. The second and third terms correspond to the intermediate divider along with the HM and the filtering of the main PLL, respectively. The first term reflects the filtering effect of the auxiliary PLL with the FFNC, whose characteristic is illustrated in Fig. 10. When the FFNC is turned off, $H_{SF}(s)$ disappears in (12), and the first term exhibits a conventional low-pass characteristic of the auxiliary PLL with its nominal low-bandwidth, as shown by the blue line in Fig. 10. In contrast, when the FFNC is turned on, the overall characteristic of this term is dominated by the $H_{SF}(s)$ with a bandwidth close to $f_{REF}/2$, as shown by the red line in Fig. 10. This indicates that the auxiliary PLL has no effective filtering on the reference-referred noise $\phi_{nEQUIVA}$ due to the FFNC. Nonetheless, this has negligible influence on the overall performance since $\phi_{nEQUIVA}$ will always experience the low-pass filtering of the main PLL, which is designed to have a much narrower bandwidth than the auxiliary PLL.

IV. CIRCUIT IMPLEMENTATION

Fig. 11 shows the implementation details of the proposed circuit. The auxiliary PLL employs an SPD and an active LF based on a feedback amplifier. A low-power LC VCO is used for the auxiliary PLL, while a ring VCO is also implemented just for verification purposes. The LC -VCO can be switched

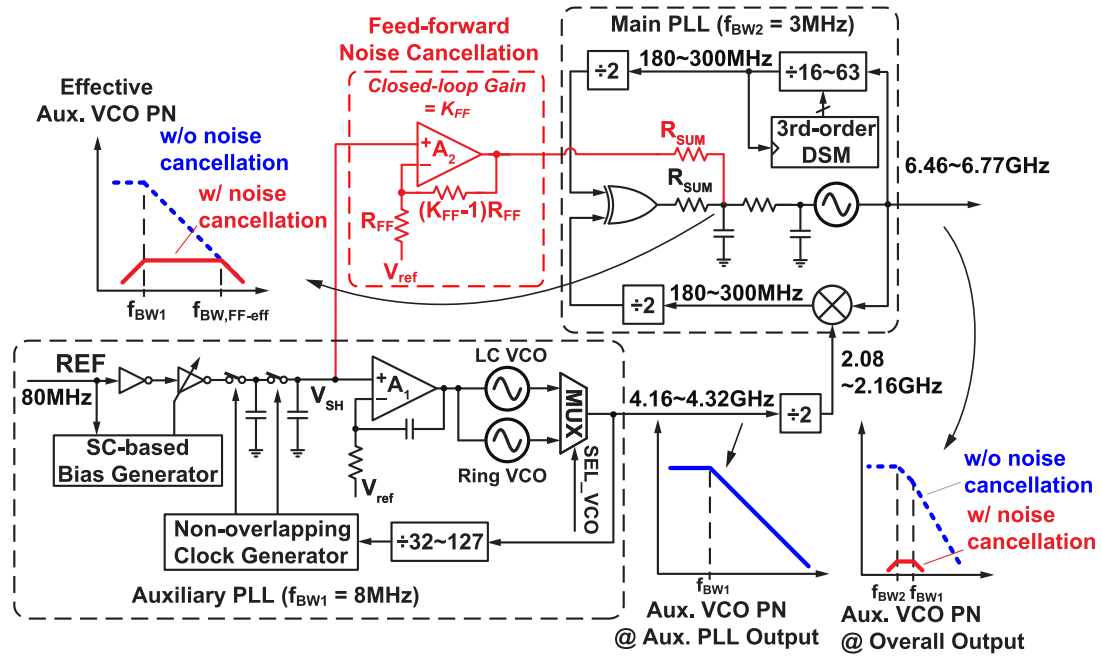


Fig. 11. Implementation details of the proposed circuit.

to a ring VCO with worse PN to further demonstrate the effect of the proposed technique. The main PLL is based on an XOR PD and a cascade of two RC filters in the LF. The SPD output from the auxiliary PLL is fed to a gain stage, which is realized by a resistor-based feed-forward amplifier, as illustrated with red in Fig. 11, and then, the resulting output is added to the output of the XOR PD in the main PLL with resistive summation. The two resistors in the feed-forward path are designed with a resistance ratio of $K_{FF}-1$ so that the gain of the feed-forward path is K_{FF} . The bandwidth of the feed-forward amplifier is around 900 MHz, much higher than the PLL bandwidth such that the feed-forward gain K_{FF} can be treated as a constant throughout the derivations in Section III. A divide-by-2 is placed between the auxiliary PLL output and the third-order HM to separate the output frequencies of the auxiliary and main VCOs so that the coupling between the two VCOs is avoided.

Before explaining the details of the circuit blocks, it is instructive to first find out what are the main contributing factors for the gain error. Based on the linear model shown in Fig. 6, the gain of the main signal path can be expressed as kK_{PDM}/M , while that of the noise cancellation path is given by $K_{FF}K_{PDA}/N_A$. The gain mismatch between these two paths can be derived based on the variation of each variable in the expressions. k , M , and N_A are frequency multiplication and division ratios and are discrete values that do not suffer from any process variation as long as the circuit is operating correctly. K_{PDM} is the gain of the XOR PD, which is directly related to the supply voltage. In practical applications, however, the regulation of supply voltages is usually managed by a separate on-chip or off-chip system such as low-dropout (LDO) regulators [23], [24] that operate independently of the PLL architecture with no impact on the settling behavior of the loop. Supposing this kind of system integration, we assume

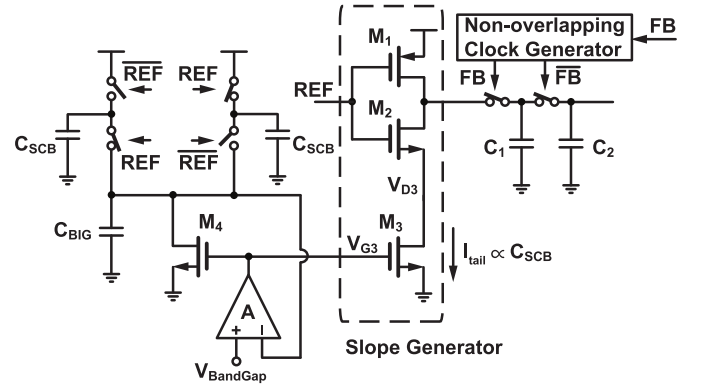


Fig. 12. SPD with switched-capacitor-based biasing.

that the variation in K_{PDM} is negligible in our analysis. This leaves K_{FF} and K_{PDA} , which are the gains of the feed-forward stage and the SPD in the auxiliary PLL, respectively. The gain of the feed-forward stage can also be realized with small process variation since it is dependent on the ratio of resistors. Therefore, the gain of the SPD is the main cause of the gain error in this architecture.

In order to realize the SPD gain with low process and temperature impact, we employ a slope generator using the switched-capacitor-based bias generator proposed in [25]. As shown in Fig. 12, the slope generator is an inverter with a tail current source, which then drives the sampling capacitor. The key point of this architecture is that the bias for the tail current source is generated using a current mirror with a switched-capacitor-based resistor. As a result, the pull-down current of the slope generator is proportional to the capacitance in the switched-capacitor-based resistor. Furthermore, the generated slope is proportional to the pull-down current divided by the sampling capacitance. As a result, the slope (and the SPD

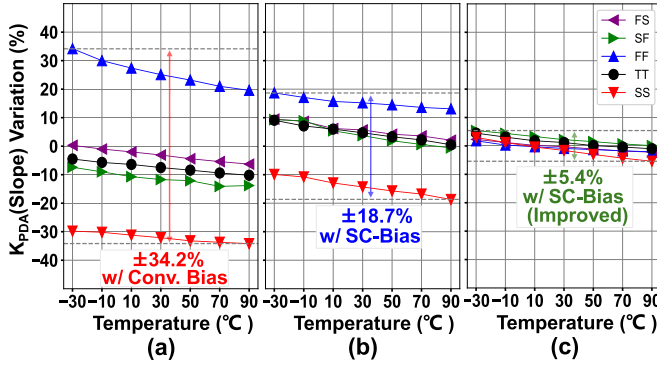


Fig. 13. Slope variation of (a) conventional SPD, (b) SC-biased SPD, and (c) improved SC-biased SPD.

gain) is proportional to the ratio of the sampling capacitance and the capacitance in the switched-capacitor-based resistor, a quantity that is relatively process invariant. In contrast, the gain of a conventional SPD with a slope generator biased by a simple current mirror and a resistor would be inversely proportional to the sampling capacitance, which is highly sensitive to process variation. Based on post-layout simulation results over different process corners and temperatures, Fig. 13(a) and (b) shows the slope variation of the conventional SPD and the implemented switched-capacitor-biased (SC-biased) SPD, respectively. The conventional SPD has a slope variation of $\pm 34.2\%$, while for the implemented SC-biased SPD, the variation is $\pm 18.7\%$, which falls within the acceptable range based on the noise calculation shown in Fig. 9 and has a negligible effect on the overall PLL performance.

It should be noted that the slope generator implemented in this chip has some suboptimal design for the transistor size of the inverter switch and the tail current source. In Fig. 12, W/L of the inverter switches M_1 and M_2 was designed too small compared to the tail current source M_3 . As a result, the voltage drops across M_1 and M_2 were excessive, causing the drain voltage of M_3 (V_{D3}) to become too low for M_3 to operate in the saturation region. Consequently, the linearity of I_{tail} with respect to C_{SCB} was reduced, degrading the PVT robustness of the SPD slope. Fig. 13(c) shows the post-layout simulation results of an improved version of the implemented SPD where the transistor sizes are re-adjusted, demonstrating that a smaller variation of $\pm 5.4\%$ can be achieved with the current structure. The simulated power and PN of the improved SPD also exhibit no significant changes compared to the implemented SPD, which means that the implemented SPD can be readily replaced with the improved one for better PVT robustness without performance degradation. The variation of the K_{FF} based on post-layout simulation is also shown in Fig. 14(a), demonstrating a variation of $\pm 9.6\%$. In addition, the variation of the product of K_{FF} and the improved K_{PDA} , which can indicate the overall FFNC gain variation, is shown in Fig. 14(b). We can see that a worst case FFNC gain error within $\pm 10.2\%$ can be achieved over a wide range of temperatures across all the process corners. As indicated in Fig. 9, this acceptably small level of the gain error is shown to cause negligible performance degradation in the overall PN. That being said, foreground calibration, as described in [20],

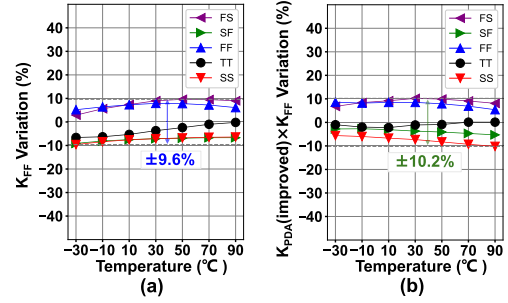


Fig. 14. Variation of (a) K_{FF} and (b) product of K_{FF} and K_{PDA} over different processes and temperatures.

can also be used to calibrate the FFNC gain and further enhance its effectiveness, such as optimizing the XOR PD supply voltage V_{DD} based on its output to minimize the gain error. However, it would yield only minimal improvement in jitter and FoM at the cost of increased design complexity, as the PN of the low-power auxiliary VCO already contributes little to the overall jitter, which will be presented in Section V.

The active LF in the auxiliary PLL serves two purposes in our architecture. It provides a dc pole so that the locking point of the SPD is fixed, reducing the voltage-dependent gain variation. It also acts as an isolation buffer so that the SPD output fed to the feed-forward path is not affected by kick-back from the auxiliary VCO. An LF exhibiting such characteristics is usually realized using a G_m cell followed by an RC network [26]. However, as shown in Fig. 15, the LF based on a feedback amplifier used in our work offers some advantages over the G_m -cell-based configuration. First, the proposed architecture provides better control of the loop characteristic over process variation. The G_m -cell-based configuration is an open-loop circuit, and its transfer function is a direct product of the LF characteristic and the transconductance of the G_m cell. A variation in the G_m cell transconductance will significantly influence the transfer function of the LF $H_{\text{LFA}}(s)$, thus affecting the loop gain of the auxiliary PLL and its bandwidth. In contrast, the proposed LF uses an amplifier in a closed-loop configuration so that the amplifier voltage gain variation has a negligible impact on the loop bandwidth. Second, the proposed configuration is easier to implement in terms of the desired output impedance. In the G_m -cell-based configuration, the LF output impedance has to be large since its output is a current, but cascode architectures with high output impedance are relatively difficult to achieve especially in advanced process nodes where the supply voltage is low. In contrast, the proposed configuration requires the amplifier to have a small output impedance since its output is a voltage, which is much easier to achieve than a high output impedance. These advantages make the proposed configuration more desirable to be used than the conventional G_m -cell-based one.

V. MEASUREMENT RESULTS

The chip photograph of the proposed PLL with the FFNC technique is shown in Fig. 16. The design was fabricated in a 65-nm CMOS process and the layout occupies an active area of 0.36 mm^2 .

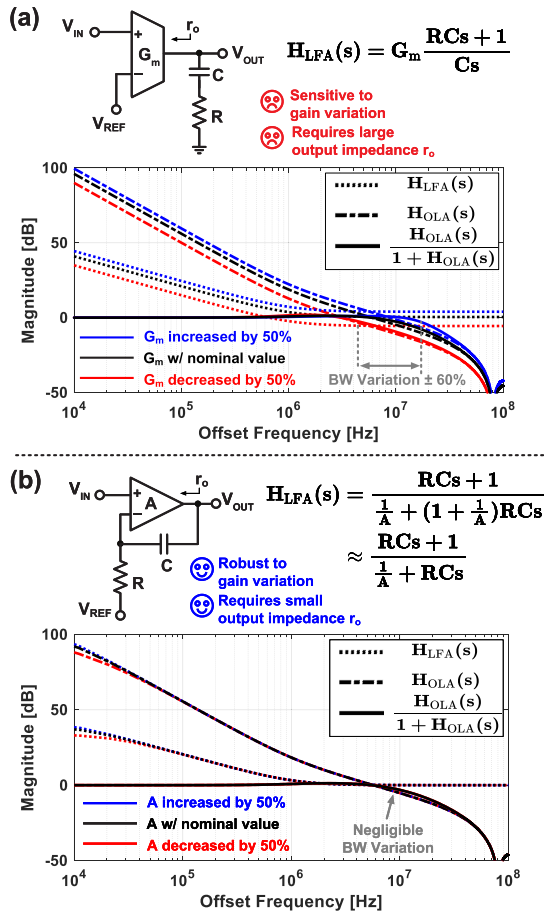


Fig. 15. (a) Conventional LF based on a G_m cell and (b) proposed LF based on a feedback amplifier and a comparison of their robustness on gain variation.

Fig. 17 shows the measured PN spectrum of the auxiliary PLL in standalone operation along with the free-running auxiliary VCO PN when we use the LC VCO and ring VCO. The signal is measured after an on-chip divide-by-2, so the PN spectrum is lowered by 6 dB. The PN of the LC VCO is -108 dBc/Hz at a 1-MHz offset, while that of the ring VCO is -86 dBc/Hz, both normalized to the output frequency. The auxiliary loop bandwidth is around 8 MHz for the LC VCO case and 12 MHz for the ring VCO case. Both values are sufficiently smaller than the reference frequency of 80 MHz, meaning that the risk of instability is properly avoided. The resulting integrated jitter values are 195 fs for the LC VCO case and 849 fs for the ring VCO case.

Fig. 18 shows the measured PN spectrum of the overall PLL with the FFNC turned on/off when we use the LC VCO and ring VCO for the auxiliary PLL. The signal is measured after an on-chip divide-by-4, so the PN spectrum is lowered by 12 dB. For both cases, the PN is greatly suppressed when the noise cancellation is enabled. For the LC VCO case, when the FFNC is turned on, the integrated jitter is reduced from 149 to 106 fs. For the ring VCO case, the integrated jitter is 392 and 128 fs when we turn off/on the noise cancellation, respectively. We can see that even though the PN contribution of the auxiliary ring VCO is very large, the resulting PN and jitter after noise cancellation are greatly lowered, even comparable

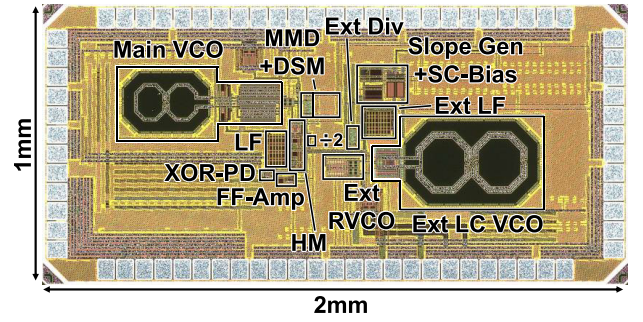


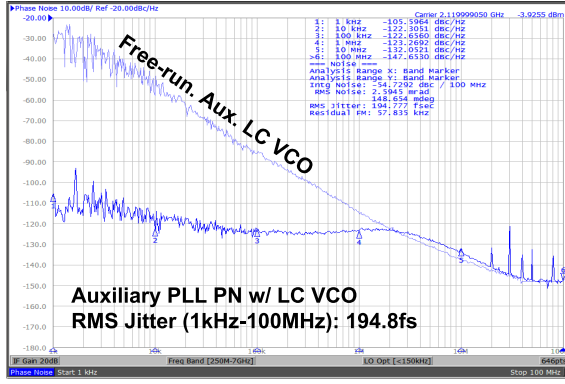
Fig. 16. Chip photograph.

to the LC VCO case. These results show the effectiveness of the proposed noise cancellation technique.

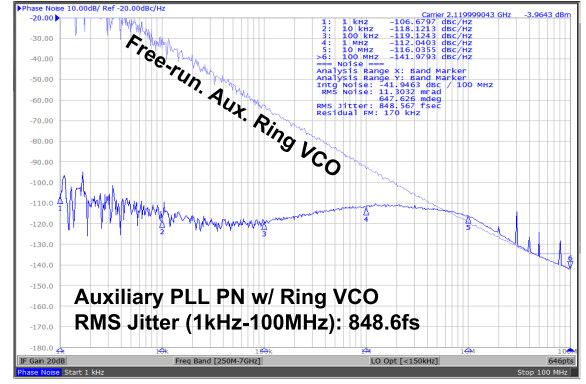
Fig. 19 shows the measured PN spectrum of the overall PLL for the LC VCO case with the noise cancellation turned on/off, along with the calculated contributions from the different circuit blocks. The measured free-running main VCO PN is also included and is -117 dBc/Hz at 1-MHz offset normalized to the output frequency. The quantization noise from the fractional divider is suppressed to a negligible level thanks to the unity dc gain provided by the HM-based dual-feedback architecture. The integrated jitter of the auxiliary VCO PN is 15.4 and 114.3 fs with and without the FFNC, respectively, and their percentage over the overall jitter in each mode is 2% and 54%, respectively. This shows that the proposed technique successfully lowers the auxiliary VCO PN contribution, which would have dominated the overall jitter without cancellation. The out-of-band PN over 20 MHz in the measurement result is likely limited by white noise from the PLL output buffer. However, this discrepancy from the calculation results contributes a negligible amount to the overall integrated jitter.

Fig. 20 shows the power consumption breakdown for the LC VCO case and the ring VCO case, respectively. We can see from Fig. 20(a) that the LC VCO in the auxiliary PLL consumes 1.5 mW (including its buffers), which is just 12% of the overall power consumption. When the ring VCO is used for the auxiliary PLL, though its power is roughly half of the entire power, this configuration is used only for verifying the impact of the proposed techniques. Based on the integrated jitter turning off/on the FFNC illustrated in Fig. 18, with a low-power overhead of 0.63 mW from the feed-forward path in both LC and ring VCO cases, the FFNC improves the PLL overall jitter-power FoM from -245.8 to -248.5 dB for the LC VCO case and from -234.8 to -244.4 dB for the ring VCO case. In contrast, suppressing the auxiliary VCO PN by directly increasing its power consumption would lead to a significant power overhead and drastically degrade the overall FoM. Based on the calculation in Fig. 19, to achieve a PN contribution as small as 2% but without using the FFNC, the auxiliary VCO PN needs to be lowered by at least 17 dB. Maintaining the same VCO FoM, the auxiliary VCO would burn a $50\times$ larger power, which is 75 mW, degrading the overall PLL FoM to -240.1 dB.

Fig. 21 shows the output spectrum containing the fractional spur and a plot of the fractional spur values for different fractional values of the FCW. Again, since the signal is

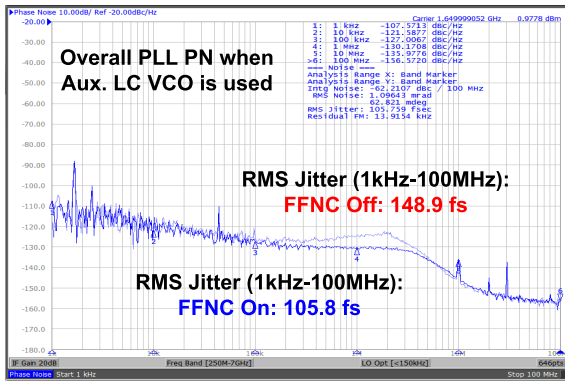


(a)

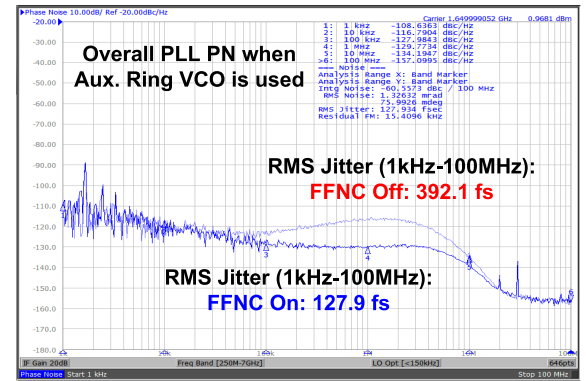


(b)

Fig. 17. Measured PN of the auxiliary PLL when (a) LC VCO is used and (b) ring VCO is used.



(a)



(b)

Fig. 18. Measured PN of the overall PLL when (a) LC VCO is used for the auxiliary PLL and (b) ring VCO is used.

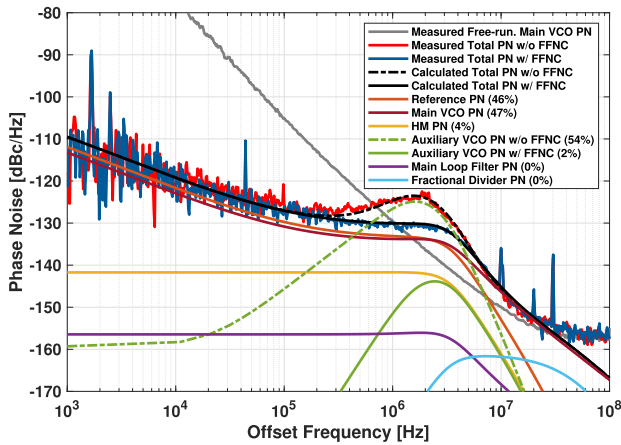


Fig. 19. Measured PN of the overall PLL using auxiliary LC VCO with the FFNC turned on and the calculated PN contribution.

measured after the on-chip divide-by-4, the measured spur values on the spectrum are 12 dB lower than the values at the actual PLL output. The worst case fractional spur value is around -63 dBc.

Fig. 22 shows the output spectrum containing the reference spur at 80 MHz and coupling spurs. The reference spurs are around -77 dBc when referred to the PLL output. The coupling spurs mainly result from the interference between the

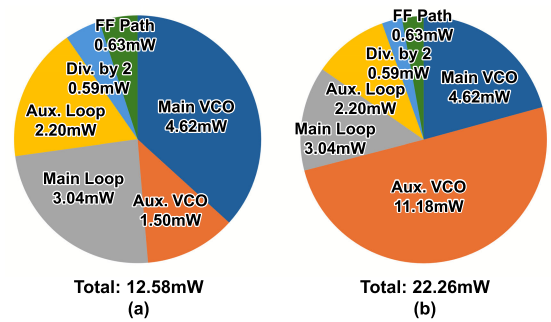


Fig. 20. Power consumption breakdown of the auxiliary PLL when (a) LC VCO is used and (b) ring VCO is used.

DSM and the reference blocks. For the measurement shown in Fig. 22, the DSM clock frequency is 244.56 MHz, while the reference frequency is 80 MHz. It is likely that the perturbation caused by the DSM is down-converted to 4.56 MHz by the third harmonic of the reference frequency resulting in the coupling spur. The coupling spur can be reduced by further improving the isolation between blocks and separating the power supplies among other things.

Table I shows a performance comparison with other state-of-the-art works. We can see that both the jitter-power performance and spurious performance of our work are competitive with the other state-of-the-art works of fractional-N

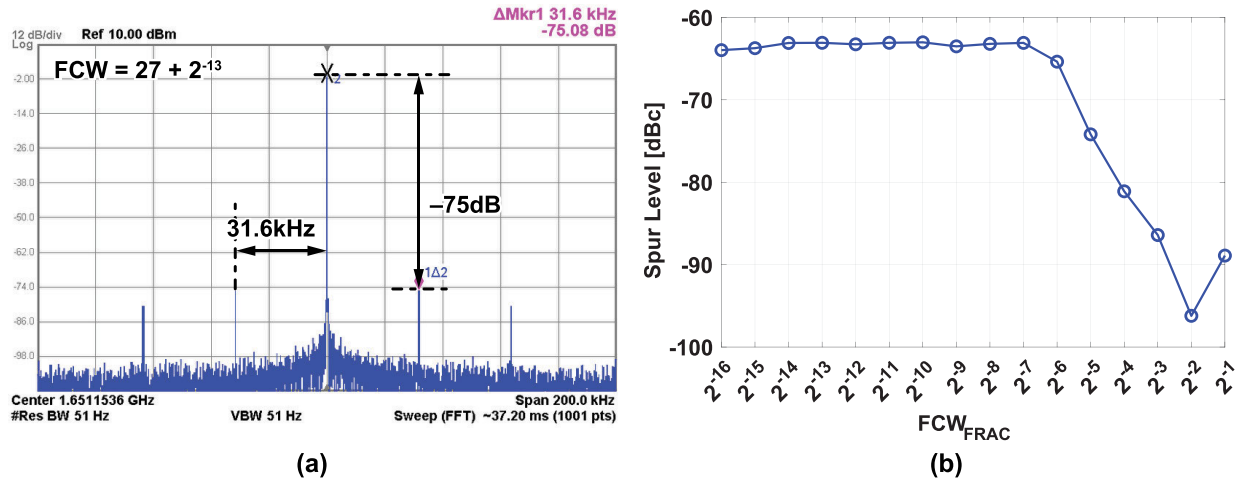


Fig. 21. (a) Output spectrum containing the fractional spur and (b) plot of the fractional spur values for different fractional values of the FCW.

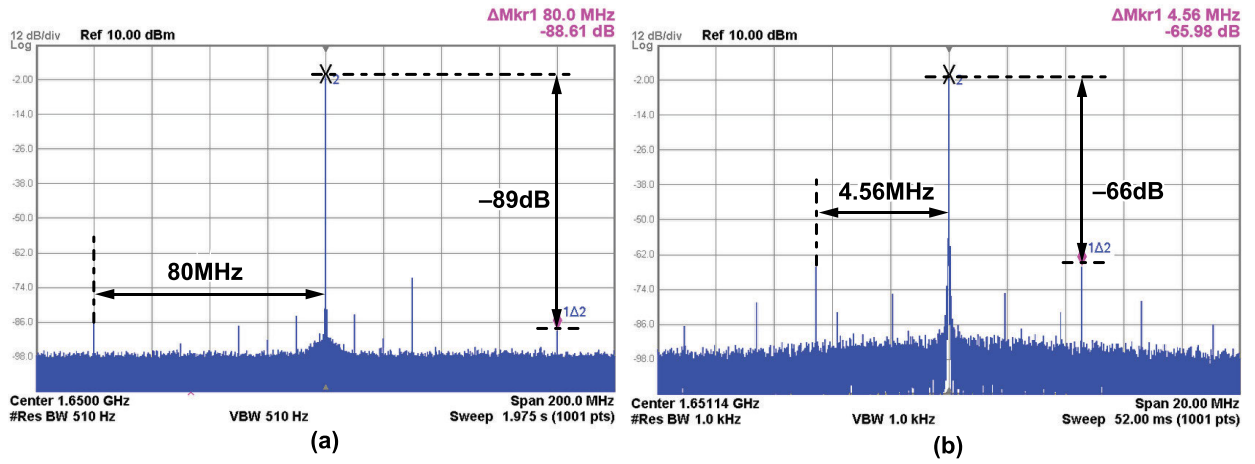


Fig. 22. (a) Output spectrum containing the reference spur at 80 MHz and (b) coupling spurs.

TABLE I
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART FRACTIONAL- N PLLs

	This Work	[12] D. Yang ISSCC'19	[5] D. Murphy JSSC'23	[27] D. Xu ISSCC'24	[4] J. Kim JSSC'24	[3] S. Dartizio ISSCC'23	[28] G. Castoro ISSCC'23	[29] Z. Gao JSSC'23	[30] Z. Xu VLSI'21
Q-noise Suppression Technique	HM-based w/ Noise Cancellation	HM-based Tripple-Loop	Overclocking DSM + Multipath PD	Cascaded Divider + Pseudo-Diff. DTC	DAC w/ SCF DPD	ICS DTC + FCW Dither	Multi-DTC	TAU + TDC	Hybrid w/ DTC + SPD
Calibration Required for Q-noise Suppression?	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Ref. Freq. [MHz]	80	60	79.96	100	100	250	250	40	50
Output Freq. [GHz]	6.5-6.8	7.0-9.0	4.66-5.22	6.5-7.5	12.8-15.0	9.25-10.5	9.25-10.75	2.56-4.1	3.3
Integrated Jitter [fs]	106 (1k-100M)	131 (10k-10M)	154 (10k-100M)	143.7 (1k-100M)	104 (10k-100M)	76.7 (10k-100M)	77 (1k-100M)	182 (10k-40M)	236 (10k-10M)
Power [mW]	12.58	13.4	1.11	8.89	7.3	17.2	17.9	3.48	4.6
FoM* [dB]	-248.5	-246.4	-255.8	-247.4	-251	-249.9	-249.7	-249.4	-246
FoM _{REF} ** [dB]	-239.5	-238.6	-246.8	-237.4	-241.0	-235.9	-235.7	-243.4	-239.0
Fractional Spur [dBc]	-63	-70	-60	-62.1	-58	-71.9	-60.3	-59	-53
Ref. Spur [dBc]	-76.7	-66	<-70	N/A	N/A	-70.5	-71.1	-73.5	-80
Process [nm]	65	16	7	65	65	28	28	40	65
Active Area [mm ²]	0.36	0.25	0.21	0.23	0.21	0.33	0.36	0.31	N/A

*FoM = $10\log((\text{Power}/1\text{mW}) \cdot (\text{Jitter}/1\text{s})^2)$

**FoM_{ref} = $10\log((\text{Power}/1\text{mW}) \cdot (\text{Jitter}/1\text{s})^2) + 10\log(f_{\text{ref}}/10\text{MHz})$

PLLs. Specifically, among the PLLs that suppress the Q-noise without requiring calibration, our work has achieved the lowest integrated jitter with the combination of the HM-based

dual-feedback architecture and the proposed FFNC technique. Meanwhile, it is worth noting that some of the works [3], [28] that employed DTC-based Q-error cancellation achieved

better jitter performance than this work. However, these works have used a higher reference frequency of 250 MHz, which is more advantageous in terms of the reduced loop gain, but less practical for the cases with a commercial XTAL as a reference. In contrast, our work utilizes a lower reference frequency of 80 MHz and, thus, outperforms these two works significantly in terms of FoM_{ref} . In addition, our work does not require background digital calibration, resulting in a shorter settling time and, therefore, greater frequency agility.

VI. CONCLUSION

In this article, we presented an HM-based dual-feedback fractional- N PLL with an FFNC technique to suppress the noise and power overhead from the auxiliary PLL. The HM-based feedback provides unity dc gain for the main PLL, effectively suppressing the quantization noise to a negligible level without digital calibration. Distinct from prior works that require a high-speed delay line with large noise and power overhead to perform the FFNC, the proposed technique feed-forwards the noise in the voltage domain with a closed-loop amplifier and, thus, significantly improves the PN of the auxiliary VCO with a low-power overhead. To maintain a small error in the feed-forward gain, we use a switched-capacitor-based bias generator to ensure the robustness of the SPD slope to PVT variations, eliminating the need for gain calibration required in the prior work with a high-speed delay line. Thanks to the proposed architecture, this work has achieved a 106-fs rms jitter and a -63 -dBc worst case fractional spur, which is in line with the state-of-the-art fractional- N PLLs.

ACKNOWLEDGMENT

The chip design in this study was supported through the activities of VLSI Design and Education Center (VDEC), The University of Tokyo, in collaboration with Cadence Design Systems, Inc., Nihon Synopsys G.K., and Siemens Electronic Design Automation Japan K.K.

REFERENCES

- [1] S. Levantino, G. Marzin, and C. Samori, "An adaptive pre-distortion technique to mitigate the DTC nonlinearity in digital PLLs," *IEEE J. Solid-State Circuits*, vol. 49, no. 8, pp. 1762–1772, Aug. 2014, doi: [10.1109/JSSC.2014.2314436](#).
- [2] W. Wu et al., "A 28-nm 75-fsrms analog fractional- N sampling PLL with a highly linear DTC incorporating background DTC gain calibration and reference clock duty cycle correction," *IEEE J. Solid-State Circuits*, vol. 54, no. 5, pp. 1254–1265, May 2019, doi: [10.1109/JSSC.2019.2899726](#).
- [3] S. M. Dartizio et al., "A low-spur and low-jitter fractional- N digital PLL based on an inverse-constant-slope DTC and FCW subtractive dithering," *IEEE J. Solid-State Circuits*, vol. 58, no. 12, pp. 3320–3337, Dec. 2023, doi: [10.1109/jssc.2023.3311681](#).
- [4] J. Kim, Y. Jo, H. Park, T. Seong, Y. Lim, and J. Choi, "A 12.8–15.0-GHz low-jitter fractional- N subsampling PLL using a voltage-domain quantization-error cancellation," *IEEE J. Solid-State Circuits*, vol. 59, no. 2, pp. 424–434, Feb. 2024, doi: [10.1109/jssc.2023.3297618](#).
- [5] D. Murphy, D. Yang, H. Darabi, and A. Behzad, "A calibration-free Fractional- N analog PLL with negligible DSM quantization noise," *IEEE J. Solid-State Circuits*, vol. 58, no. 9, pp. 1–13, Sep. 2023, doi: [10.1109/JSSC.2023.3263075](#).
- [6] L. Kong and B. Razavi, "A 2.4-GHz 6.4-mW fractional- N inductorless RF synthesizer," *IEEE J. Solid-State Circuits*, vol. 52, no. 8, pp. 2117–2127, Aug. 2017, doi: [10.1109/JSSC.2017.2686838](#).
- [7] D. Park and S. Cho, "A 14.2 mW 2.55-to-3GHz cascaded PLL with reference injection, 800MHz delta-sigma modulator and 255fsrms integrated jitter in 0.13 μ m CMOS," in *Proc. IEEE Int. Solid-State Circuits Conf.*, San Francisco, CA, USA, Feb. 2012, pp. 344–346, doi: [10.1109/ISSCC.2012.6177038](#).
- [8] F. Song, Y. Zhao, B. Wu, L. Tang, L. Lin, and B. Razavi, "16.5 A Fractional- N synthesizer with 110fsrms jitter and a reference quadrupler for wideband 802.11ax," in *Proc. IEEE Int. Solid-State Circuits Conf.*, San Francisco, CA, USA, Feb. 2019, pp. 264–266, doi: [10.1109/ISSCC.2019.8662488](#).
- [9] M.-S. Yuan, C.-C. Li, C.-C. Liao, Y.-T. Lin, C.-H. Chang, and R. B. Staszewski, "A 0.45 V sub-mW all-digital PLL in 16nm FinFET for Bluetooth low-energy (BLE) modulation and instantaneous channel hopping using 32.768 kHz reference," in *Proc. IEEE Int. Solid-State Circuits Conf.*, San Francisco, CA, USA, Feb. 2018, pp. 448–450, doi: [10.1109/ISSCC.2018.8310377](#).
- [10] N. Mousavi, Z. Wang, D. Cabric, and R. Harjani, "A 0.4–1.0 GHz, 47 MHz/s frequency-hopped TXR front end with 20 dB in-band blocker rejection," *IEEE J. Solid-State Circuits*, vol. 54, no. 7, pp. 1917–1928, Jul. 2019, doi: [10.1109/JSSC.2019.2911407](#).
- [11] J. Prinzie et al., "A fast locking 5.8–7.2-GHz fractional- N synthesizer with sub-2- μ s settling in 22-nm FDSOI," *IEEE Solid-State Circuits Lett.*, vol. 3, pp. 546–549, 2020, doi: [10.1109/LSSC.2020.3036122](#).
- [12] D. Yang et al., "16.6 a calibration-free triple-loop bang-bang PLL achieving 131fsrms jitter and-70dBc fractional spurs," in *Proc. IEEE Int. Solid-State Circuits Conf.*, San Francisco, CA, USA, Feb. 2019, pp. 266–268, doi: [10.1109/ISSCC.2019.8662494](#).
- [13] M. Osada, Z. Xu, and T. Iizuka, "A 3.2-to-3.8 GHz harmonic-mixer-based dual-feedback fractional- N PLL achieving -65 dBc in-band fractional spur," *IEEE Solid-State Circuits Lett.*, vol. 3, pp. 534–537, 2020, doi: [10.1109/LSSC.2020.3037311](#).
- [14] D. Yang, D. Murphy, H. Darabi, A. Behzad, R. Ruby, and R. Parker, "An FBAR driven -261 dB FOM fractional- N PLL," in *Proc. IEEE Radio Freq. Integr. Circuits Symp. (RFIC)*, Atlanta, GA, USA, Jun. 2021, pp. 147–150, doi: [10.1109/RFIC51843.2021.9490409](#).
- [15] D. Yang et al., "A harmonic-mixing PLL architecture for millimeter-wave application," *IEEE J. Solid-State Circuits*, vol. 57, no. 12, pp. 3552–3566, Dec. 2022, doi: [10.1109/JSSC.2022.3209614](#).
- [16] M. Osada, Z. Xu, Z. Yang, and T. Iizuka, "A fractional- N ring PLL using harmonic-mixer-based dual-feedback and split-feedback frequency division with phase-domain filtering," *IEEE J. Solid-State Circuits*, vol. 59, no. 7, pp. 2171–2184, Jul. 2024, doi: [10.1109/jssc.2024.3353219](#).
- [17] F. Bruccoleri, E. A. M. Klumperink, and B. Nauta, "Wide-band CMOS low-noise amplifier exploiting thermal noise canceling," *IEEE J. Solid-State Circuits*, vol. 39, no. 2, pp. 275–282, Feb. 2004, doi: [10.1109/JSSC.2003.821786](#).
- [18] D. Murphy et al., "A blocker-tolerant, noise-cancelling receiver suitable for wideband wireless applications," *IEEE J. Solid-State Circuits*, vol. 47, no. 12, pp. 2943–2963, Dec. 2012, doi: [10.1109/JSSC.2012.2217832](#).
- [19] A. Homayoun and B. Razavi, "On the stability of charge-pump phase-locked loops," *IEEE Trans. Circuits Syst. I: Reg. Papers*, vol. 63, no. 6, pp. 741–750, Jun. 2016, doi: [10.1109/TCSI.2016.2537823](#).
- [20] S. S. Nagam and P. R. Kinget, "A low-jitter ring-oscillator phase-locked loop using feedforward noise cancellation with a sub-sampling phase detector," *IEEE J. Solid-State Circuits*, vol. 53, no. 3, pp. 703–714, Mar. 2018, doi: [10.1109/JSSC.2017.2788876](#).
- [21] Y. Zhao, M. Forghani, and B. Razavi, "A 20-GHz PLL with 20.9-fs random jitter," *IEEE J. Solid-State Circuits*, vol. 58, no. 6, pp. 1597–1609, Jun. 2023, doi: [10.1109/JSSC.2022.3225105](#).
- [22] L. Kong and B. Razavi, "A 2.4 GHz 4 mW integer- N inductorless RF synthesizer," *IEEE J. Solid-State Circuits*, vol. 51, no. 3, pp. 626–635, Mar. 2016, doi: [10.1109/JSSC.2015.2511157](#).
- [23] C.-H. Huang and W.-C. Liao, "A high-performance LDO regulator enabling low-power SoC with voltage scaling approaches," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 28, no. 5, pp. 1141–1149, May 2020, doi: [10.1109/tvlsi.2020.2972904](#).
- [24] F. U. Ahmed, Z. T. Sandhie, L. Ali, and M. H. Chowdhury, "A brief overview of on-chip voltage regulation in high-performance and high-density integrated circuits," *IEEE Access*, vol. 9, pp. 813–826, 2021, doi: [10.1109/ACCESS.2020.3047347](#).
- [25] J. Jung et al., "A 2.4-to-4.2GHz 440.2fsrms-integrated-jitter 4.3 mW ring-oscillator-based PLL using a switched-capacitor-bias-based sampling PD in 4nm FinFET CMOS," in *Proc. IEEE Symp. VLSI Technol. Circuits*, Kyoto, Japan, Jun. 2023, pp. 1–2, doi: [10.23919/vlsitechnologyandcir57934.2023.10185300](#).

- [26] G. Jin, F. Feng, W. Chen, Y. Shu, X. Luo, and X. Gao, "A fractional- N sampling PLL with a merged constant-slope DTC and sampling PD," *IEEE J. Solid-State Circuits*, vol. 59, no. 8, pp. 2407–2417, Aug. 2024, doi: [10.1109/jssc.2024.3358564](https://doi.org/10.1109/jssc.2024.3358564).
- [27] D. Xu et al., "10.3 a 7GHz digital PLL with cascaded fractional divider and pseudo-differential DTC achieving -62.1dBc fractional spur and 143.7fs integrated jitter," in *Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC)*, San Francisco, CA, USA, Feb. 2024, pp. 192–194, doi: [10.1109/isscc49657.2024.10454284](https://doi.org/10.1109/isscc49657.2024.10454284).
- [28] G. Castoro et al., "4.5 A 9.25GHz digital PLL with fractional-spur cancellation based on a multi-DTC topology," in *Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC)*, Feb. 2023, pp. 82–84, doi: [10.1109/isscc42615.2023.10067351](https://doi.org/10.1109/isscc42615.2023.10067351).
- [29] Z. Gao et al., "A low-spur fractional- N PLL based on a time-mode arithmetic unit," *IEEE J. Solid-State Circuits*, vol. 58, no. 6, pp. 1552–1571, Jun. 2023, doi: [10.1109/JSSC.2022.3209338](https://doi.org/10.1109/JSSC.2022.3209338).
- [30] Z. Xu, M. Osada, and T. Iizuka, "A 3.3-GHz 4.6-mW fractional- N type-II hybrid switched-capacitor sampling PLL using CDAC-embedded digital integral path with -80-dBc reference spur," in *Proc. Symp. VLSI Circuits*, Kyoto, Japan, Jun. 2021, pp. 1–2, doi: [10.23919/VLSICircuits52068.2021.9492381](https://doi.org/10.23919/VLSICircuits52068.2021.9492381).
- [31] S. Levantino, "Calibration techniques in phased-locked loops: Theoretical basis and practical applications," *IEEE Solid State Circuits Mag.*, vol. 16, no. 3, pp. 37–44, May 2024, doi: [10.1109/mssc.2024.3408714](https://doi.org/10.1109/mssc.2024.3408714).
- [32] G. Marzin, S. Levantino, C. Samori, and A. L. Lacaita, "2.9 A background calibration technique to control bandwidth in digital PLLs," in *Proc. IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers (ISSCC)*, San Francisco, CA, USA, Feb. 2014, pp. 54–55, doi: [10.1109/ISSCC.2014.6757335](https://doi.org/10.1109/ISSCC.2014.6757335).
- [33] A. Elshazly, R. Inti, W. Yin, B. Young, and P. K. Hanumolu, "A 0.4-to-3GHz digital PLL with supply-noise cancellation using deterministic background calibration," in *Proc. May IEEE Int. Solid-State Circuits Conf.*, 2011, pp. 92–94, doi: [10.1109/ISSCC.2011.5746233](https://doi.org/10.1109/ISSCC.2011.5746233).



Yuyang Zhu (Graduate Student Member, IEEE) received the B.S. degree in opto-electronic information science and engineering from Sun Yat-sen University, Guangzhou, China, in 2021, and the M.S. degree in electrical engineering from The University of Tokyo, Tokyo, Japan, in 2024, where he is currently pursuing the Ph.D. degree.

His current research interests include high-performance inductorless frequency synthesizers for wireless and wireline communications.



Haoming Zhang (Graduate Student Member, IEEE) received the B.E. degree in integrated circuits and systems from Dalian University of Technology, Dalian, China, in 2021, and the M.E. degree in electrical engineering and information systems from The University of Tokyo, Tokyo, Japan, in 2023, where he is currently pursuing the Ph.D. degree with the Department of Electrical Engineering and Information Systems.

His current interests include characterization and design automation of integrated circuits and high-performance frequency synthesizers for wireless and wireline communications.



Masaru Osada (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from The University of Tokyo, Tokyo, Japan, in 2019, 2021, and 2024, respectively.

He is currently with Apple, Inc., as an Analog Design Engineer. His current research interests include high-performance frequency synthesizers for wireless and wireline communications.



Tetsuya Iizuka (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronic engineering from The University of Tokyo, Tokyo, Japan, in 2002, 2004, and 2007, respectively.

From 2007 to 2009, he was with THine Electronics Inc., Tokyo, as a High-Speed Serial Interface Circuit Engineer. He joined The University of Tokyo in 2009, where he is currently an Associate Professor with the Systems Design Laboratory, School of Engineering. From 2013 to 2015, he was a Visiting Scholar with the University of California at

Los Angeles, Los Angeles, CA, USA. His current research interests include data conversion techniques, high-speed analog integrated circuits, digitally assisted analog circuits, and VLSI computer-aided designs.

Dr. Iizuka was a member of the IEEE International Solid-State Circuits Conference (ISSCC) Technical Program Committee from 2013 to 2017 and a member of the IEEE Custom Integrated Circuits Conference (CICC) Technical Program Committee from 2014 to 2019. He is a member of the Institute of Electronics, Information and Communication Engineers (IEICE). He is currently serving as a member for the IEEE Asian Solid-State Circuits Conference (A-SSCC) and IEEE VLSI Symposium on Circuits Technical Program Committees. He was a recipient of the 21st Marubun Research Encouragement Commendation from the Marubun Research Promotion Foundation in 2018, the 13th Wakashachi Encouragement Award First Prize in 2019, and the 18th Funai Academic Prize from the Funai Foundation for Information Technology in 2019. He was a co-recipient of the IEEE International Test Conference Ned Kornfield Best Paper Award in 2016. From 2016 to 2018, he served as an Editor for *IEICE Electronics Express* (ELEX).